

Research domains in IoT

Due to the ever increasing deployment of IoT in the government and in the private sector, there is a need to analyse various factors that can affect its overall performance and efficiency.

The following are some of the key research issues in IoT:

- Security, privacy and trust architectures
- Big Data analysis and scalability
- Device interoperability and compliance
- Robustness and fault tolerance
- Cognitive networking
- Energy aware approaches
- Virtualisation
- Ontology models

Free and open source tools for IoT programming

Featured below are some well known open source tools used for IoT programming.

OpenIoT (<http://www.openiot.eu/>): This is a free and open source platform to manage and program the sensors on the cloud and the Internet based environment. The concept of Sensing-as-a-Service is being adopted in OpenIoT.

Zetta (<http://www.zettajs.org/>): This is a free and open source platform that is based on Node.js. Zetta is used to create the IoT servers that can control and run the globally distributed systems, sensors and computers, including those that are on the cloud.

DSA (<http://www.iot-dsa.org/>): Distributed Services Architecture is a powerful, open source IoT library that is available for free distribution. It enhances the performance of inter-object communication and makes it very effective. DSA provides the toolkit for managing IoT based applications, services and objects.

Node-RED (<http://nodered.org/>): This provides the programming interface and APIs for the Internet of Things. Using Node-RED, the flow based creation of remote IoT objects can be done easily with a Web browser based flow editor. In the flow editor of Node-RED, the JavaScript code can be executed and remote objects can be programmed easily with powerful functionalities.

IoTivity (<https://www.iotivity.org/>): This is a powerful open source library which enables inter-object connectivity with enormous speed and performance. It is written and programmed in C and C++. Most of the performance aware protocols (like ANT+, Bluetooth low energy, Wi-Fi Direct, Zigbee and Z-Wave) can be easily integrated with IoTivity.

There are other open source implementations for the Internet of Things as listed below.

Development toolkits and libraries

- Arduino
- Eclipse IoT Project
- Kinoma

- M2MLabs Mainspring
- Node-RED
- ThingBox

Automation for home and offices

- Eclipse SmartHome
- Home Gateway Initiative (HGI)
- Ninja Blocks
- openHAB
- PrivateEyePi
- RaZberry
- The Thing System

Middleware

- IoTSys
- Kaa
- OpenIoT
- OpenRemote

Operating systems

- AllJoyn
- Brillo
- Contiki
- FreeRTOS
- Raspbian
- RIOT
- Spark
- TinyOS

IoT integration tools and horizontal platforms

- Canopy
- Chimera IoT
- DeviceHive
- IoT Toolkit
- M2MLabs Mainspring
- Mango
- Nimbits
- Open Source Internet of Things (OSIoT)
- OpenRemote
- Pico Labs
- prpl Foundation
- RabbitMQ
- SiteWhere
- SiteWhere
- ThingSpeak
- webinos
- Yaler

Protocols

- Advanced Message Queuing Protocol (AMQP)
- Constrained Application Protocol (CoAP)
- Extensible Messaging and Presence Protocol (XMPP)
- OASIS Message Queuing Telemetry Transport (MQTT)
- Very Simple Control Protocol (VSCP)